The Age of Industrialisation

The big idea

We usually picture industrialisation as giant factories, smoking chimneys and steam engines. But this chapter shows that industrial change was slow and uneven. Even before factories, there was large-scale production for international markets. Hand labour remained important for a long time, and in colonies like India, industrialisation took a different path shaped by colonial rule.

Before the Industrial Revolution

PROTO-INDUSTRIALISATION

A phase of **large-scale industrial production for an international market**, which was **not based on factories**. It existed in England and Europe even before factories began.

In the seventeenth and eighteenth centuries, merchants from towns moved to the **countryside**, supplying money to peasants and artisans to produce for international markets. Why? In towns, powerful **trade and craft guilds** (associations of producers that trained craftspeople, controlled production, regulated competition and prices) and rulers restricted the entry of new people into trade. Poor cottagers in the countryside, who no longer had access to the common lands, welcomed the extra income.

Under this system, a close relationship developed between town and countryside. A merchant clothier in England bought wool from a **wool stapler**, carried it to spinners, then weavers, **fullers** and dyers; finishing was done in London before export. Proto-industrial workers were controlled by merchants but the family farm remained the base.

The coming up of the factory

- The earliest factories in England came up by the **1730s**, but their number multiplied only in the late eighteenth century.
- **Cotton** was the first symbol of the new era. Its production boomed: in **1760**, Britain imported **2.5 million pounds** of raw cotton; by **1787** this had soared to **22 million pounds**.
- **Richard Arkwright** created the **cotton mill**: costly new machines were set up and managed under one roof, allowing careful supervision of the production process, quality control and regulation of labour.
- **James Watt** improved the steam engine produced by **Newcomen** and patented the new engine in **1781**; his industrialist friend **Matthew Boulton** manufactured the new model.
- By the **1840s**, cotton was the leading sector; after that, **iron and steel** led the way with railway expansion.

"FACTORIES REPLACED TRADITIONAL INDUSTRY OVERNIGHT"

Not true. Even in Britain, the most dynamic industries (cotton and metals) accounted for less than a fifth of workers. **Technological change was slow**: new technology was expensive and machines often broke down. Ordinary and small innovations in non-mechanised sectors (food processing, building, pottery, glass, furniture) were the basis of growth.

Hand labour and steam power

In Victorian Britain, there was **no shortage of human labour**. Poor peasants and vagrants moved to cities in large numbers in search of jobs. With plenty of labour available at low wages, industrialists had no problem of labour shortage or high wage costs. They did not want machines that needed huge capital investment.

- In many industries the demand for labour was **seasonal** (gas works, breweries, book-binders, printers, shipyards at certain seasons). Industrialists preferred hand labour, employing workers for the season.

- A range of products could be produced only with hand labour; machines produced uniform, standardised goods for a mass market. In Victorian Britain, the **upper classes** (aristocrats and bourgeoisie) **preferred things produced by hand**, symbolising refinement and class; handmade products were better finished and individually designed.
- In countries with labour shortages, like the **USA**, industrialists were keen on mechanised power.

Life of the workers. Getting a job depended on networks of friendship and kin. Many job-seekers had to wait weeks, sleeping under bridges or in night shelters. Work was seasonal, and wages rose in the early nineteenth century but prices rose faster, so real wages fell. When the **Spinning Jenny** (devised by **James Hargreaves in 1764**) was introduced in the woollen industry, women who survived on hand spinning **attacked the new machines**, fearing unemployment. After the 1840s, building activity (roads, railway stations, tunnels) opened up new jobs.

Industrialisation in the colonies

The age of Indian textiles. Before machine industries, **silk and cotton goods from India dominated the international textile market**. Armenian and Persian merchants took goods from Punjab to Afghanistan, eastern Persia and Central Asia. Pre-colonial ports included **Surat** on the Gujarat coast (linking to the Gulf and Red Sea ports), **Masulipatnam** on the Coromandel coast, and **Hoogly** in Bengal (trade links with Southeast Asian ports).

By the 1750s this network was breaking down as European companies gained power. **Bombay and Calcutta** grew as new ports, under colonial control.

What happened to weavers? After the East India Company established political power (after the 1760s and 1770s), it set out to eliminate competition and gain direct control over weavers:

- It appointed paid servants called **gomasthas** to supervise weavers, collect supplies and examine the quality of cloth.

- It prevented weavers from dealing with other buyers through a system of **advances**; weavers who took loans had to hand over cloth to the gomastha.
- Gomasthas were outsiders who acted arrogantly and punished weavers for delays, often by flogging. In many places in **Carnatic and Bengal**, weavers deserted villages, revolted along with village traders, or refused loans and took up agriculture.

Manchester comes to India. In 1772, Henry Patullo, a Company official, said demand for Indian textiles could never reduce. But by the beginning of the nineteenth century, exports of Indian textiles declined. Manchester goods flooded Indian markets: by **1850**, cotton piece-goods formed over **31 per cent** of the value of Indian imports; by the **1870s**, over **50 per cent**. Weavers faced two problems: their **export market collapsed** and the **local market shrank**. By the 1860s, they also faced a **shortage of raw cotton**, because the American Civil War cut off US cotton supplies to Britain, and Britain turned to India.

Factories come up in India

Year	Factory milestone
1854	First cotton mill in **Bombay** (production began 1856)
1855	First **jute mill** in Bengal (Rishra)
1860s	**Elgin Mill** at Kanpur
1861	First cotton mill of **Ahmedabad**
1874	First spinning and weaving mill of **Madras**
1912	**J.N. Tata** set up the first iron and steel plant in India at **Jamshedpur**
1917	**Seth Hukumchand**, a Marwari businessman, set up the first Indian jute mill in Calcutta

Early entrepreneurs made money through trade with China (opium) and other ventures: **Dwarkanath Tagore** in Bengal; **Dinshaw Petit** and **Jamsetjee**

Nusserwanjee Tata in Bombay. Business groups like the **Birlas** traded with China too. Many were squeezed out by European Managing Agencies (Bird Heiglers & Co., Andrew Yule, Jardine Skinner & Co.), which mobilised capital and controlled companies.

Where did workers come from? Most came from neighbouring districts; e.g., over 50 per cent of Bombay mill workers in 1911 came from **Ratnagiri**. Industrialists usually employed a **jobber** to get new recruits. The jobber was an old and trusted worker who got people from his village, helped them settle and gave them money in times of crisis, but soon demanded money and gifts and controlled their lives.

Peculiarities of industrial growth. Early Indian mills produced **coarse cotton yarn** rather than fabric, exported to China. With the **Swadeshi movement**, nationalists mobilised people to boycott foreign cloth. During the **First World War**, Manchester imports declined and Indian mills supplied war needs (jute bags, army uniforms, tents, leather boots), so new factories came up. After the war, Manchester could never recapture its old position in India.

Small-scale industries predominate. Handloom production actually expanded in the twentieth century, helped by new technology like the **fly shuttle** (a mechanical device used for weaving, moved by ropes and pulleys, that increased productivity and made wide cloth possible). By the 1940s, over 35 per cent of handlooms were fitted with fly shuttles in Travancore, Madras, Mysore, Cochin and Bengal.

Market for goods

Producers used **advertisements** to make products appear necessary and desirable.

- Manchester industrialists put **labels** on cloth bundles; "**Made in Manchester**" in bold indicated quality.
- Labels carried **images of Indian gods and goddesses** (Krishna, Saraswati), to give divine approval and make goods familiar to buyers.

- **Calendars** were printed to popularise products: unlike newspapers and magazines, calendars were used even by people who could not read; they hung in tea shops and homes all year.
- Figures of **emperors and nawabs** suggested royal quality.
- Indian producers used nationalist messages: "if you care for the nation, buy products that Indians produce" (**Swadeshi**).

BOARD FAVOURITES

"Why did merchants move to the countryside in the 17th–18th centuries?", "Why did industrialists in Victorian Britain prefer hand labour?" (3 marks), "Role of gomasthas", "Problems faced by Indian weavers", "How did new consumers get created through advertisements?" (5 marks). Learn the factory-year table well.

Quick recap

- Proto-industrialisation: large-scale production for international markets, not in factories, controlled by merchants.
- Cotton led the new era (2.5 million lb in 1760 to 22 million lb in 1787); Arkwright's mill; Watt's engine patented 1781.
- Hand labour stayed common: cheap labour, seasonal demand, upper-class preference for handmade goods.
- Company control through gomasthas and advances ruined weavers; Manchester imports over 50% of Indian imports by the 1870s.
- First Bombay cotton mill 1854; jute mill 1855; Tata steel 1912. WWI and Swadeshi helped Indian industry.
- Labels, calendars, gods and nationalist messages created markets.

Practice questions

1. Proto-industrialisation refers to

   (a) production in large steam-powered factories (b) production only for local markets (c) large-scale production for international markets not based on factories (d) government-owned industries

2. Who devised the Spinning Jenny in 1764?

   (a) Richard Arkwright (b) James Hargreaves (c) James Watt (d) Matthew Boulton

3. Gomasthas were

   (a) Indian mill owners in Bombay (b) weavers' guild leaders (c) Manchester cloth traders (d) paid servants of the East India Company who supervised weavers

4. The first cotton mill in Bombay came up in

   (a) 1854 (b) 1861 (c) 1874 (d) 1917

5. Why did the upper classes in Victorian Britain prefer handmade goods?

   (a) They were cheaper than machine goods (b) Machines were banned in Britain (c) Handmade goods symbolised refinement and class and were individually designed (d) Hand labour was scarce

6. A jobber in Indian factories was

   (a) a government labour inspector (b) a type of loom (c) a British managing agent (d) an old, trusted worker who recruited and controlled labour from his village

7. Which ports were important for Indian trade in the pre-colonial period?

   (a) Bombay, Calcutta, Madras (b) Surat, Masulipatnam, Hoogly (c) Kochi, Kandla, Paradip (d) Chittagong, Karachi, Colombo

8. Why did industrialists in Victorian Britain prefer hand labour to machines?

9. How did the East India Company establish control over Indian weavers, and how did weavers respond?

Answers: 1. c 2. b 3. d 4. a 5. c 6. d 7. b 8. see notes 9. see notes